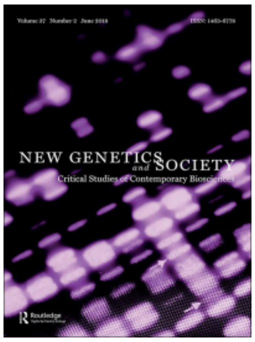

New Genetics and Society

Critical Studies of Contemporary Biosciences

ISSN: 1463-6778 (Print) 1469-9915 (Online) Journal homepage: www.tandfonline.com/journals/cngs20

Spaces of speech and places of performance: an outline of a geography of science approach to embryonic stem cell research and diabetes

Steven P. Wainwright & Clare Williams

New Genetics and Society
Vol. 27, No. 2, June 2008, 161–173

Routledge
Taylor & Francis Group

Spaces of speech and places of performance: an outline of a geography of science approach to embryonic stem cell research and diabetes

Steven P. Wainwright* and Clare Williams

Centre for Biomedicine & Society (CBAS), King's College London, University of London, UK

In this paper we develop a geography of science framework to examine the social, scientific and medical dimensions of human embryonic stem cell research. We outline David Livingstone's approach to geographies of science as "sites of speech and locations of locution" to explore the spatial shaping of science and the scientific shaping of space. Drawing upon our ongoing research on the problems and prospects of stem cell science, particularly the interactions between the bench and the bedside in the field of diabetes, we examine the influence of seminal papers on laboratory and clinical practices, and the subsequent transformation of the spaces of science on several spatial scales. We investigate the laboratory production of scientific knowledge, particularly how scientists choose which research to pursue in "scientific-landscapes-in-the-making". Finally, we explore how diverse disciplinary spaces deconstruct the stem cell transplant approach to diabetes. In conclusion, we argue that this "places of performance" perspective proffers a productive approach to understanding the shaping of contested "embryonic landscapes".

Keywords: geography; science studies; embryonic stem cells; diabetes

Introduction

Science is often viewed as producing universal knowledge and so a geographical approach to science, with an emphasis on how space shapes science, may seem an unpromising approach. In the past decade or so, however, there has been an efflorescence of studies exploring "geographies of science" (see Powell 2007). We recognize, of course, that some social scientists may dismiss the claim that "geography matters". On the one hand, some contend that "everything that is social" is reduced to an all encompassing "spatial fetishism", while others maintain the opposite arguing that place is an accretion that explains nothing that is not already adequately accounted for by the study of "the social" (such polarized debates are at the heart of the disciplinary history of geography; see Johnstone and Sideway 2004). Our position is that the social and the spatial are mutually constituted, so where you live makes a difference to the life

*Corresponding author. Email: steven.wainwright@kcl.ac.uk

ISSN 1463-6778 print/ISSN 1469-9915 online

DOI: 10.1080/14636770802077116
http://www.informaworld.com

you live. This insight also applies to science, where the type of science you produce and how you view the scientific literature is influenced by the physical place and intellectual space that you inhabit as a scientist (Shapin 1998).

Science and technology studies (STS) analysis has tended to privilege social and cultural explanatory categories over those of space and place when addressing how knowledge production and artifacts are understood differently by different groups in different places through varying practices (Henke and Gieryn 2008). We accept that there are traditions within STS that include "the spatial" (e.g. Latour 1988). Indeed, some of our previous stem cell papers use the STS "spatial concepts" of boundary-work (see Wainwright *et al.* 2006a, 2007) and boundary-objects (see Williams *et al.* forthcoming). However, it is important to note that in all of this work notions of space and place are implicit and in the background, rather than explicit and in the foreground. Hence our aim in this paper is to make place and space central to our analysis. Moreover, we seek to bring together two elements of the fields of STS and geography that have had little connection with each other, STS laboratory ethnographies and historical geographies of science, by applying some ideas developed within the geography of science tradition to our empirical data on the interactions between the bench and the bedside. Before we do this, however, we set the scene with a brief summary of two recent reviews of: (1) science studies literature that addresses place (Henke and Gieryn 2008); and (2) the geographies of science literature (Powell 2007).

A recent influential review of science studies literature that examines issues of place outlines five waves of STS research on place and science (Henke and Gieryn 2008). First, the "positivist view from nowhere" ignored place, as the production of scientific knowledge was seen as abstract and universal (e.g. Popper 1959). Second, a series of classic STS ethnographies of laboratories emphasized the context-dependency of science (e.g. Latour and Woolgar 1986). Third, historical case studies investigated how what counted as legitimate knowledge was reflected in the sites of science (Shapin and Shaffer 1985). Fourth, actor-network theory (ANT) emphasized how both human and material actors flowed through networks connecting centers of scientific calculation (Latour 1988). The fifth wave focuses on how geographical perspectives can enhance understanding of science in society by exploring three themes: why science is concentrated in particular places, how architecture affects scientific practice, and how place creates opportunities to resist the authority of science (e.g. Gieryn 1999).

A similar emphasis on the variety of approaches adopted to studying science and place is also a feature of Powell's (2007, p. 309) recent review of the geographies of science literature, where he argues that "different geographies of science are emerging" within science studies, history and geography. Powell argues that one of the main ways in which geographers have engaged with science studies is through using STS approaches, particularly ANT, as a way to map the development of aspects of geography itself (e.g. the development of economic geography; see Barnes 2001). However, he believes the major contribution by geographers to STS is through the development of historical geographies of science, which is the pre-eminent approach within the emerging sub-discipline of geography of science. In contrast, Powell states that "laboratory ethnographies have received the least consideration from geographers of science" (2007, p. 317) and yet these case studies of lab culture are one of the defining approaches and achievements of STS. Hence our paper seeks to enrich the interrelated fields of STS and geography of science, through a response to both Powell's (2007) and Livingstone's (2002, p. 85) recent calls for research on "the cultural geographies of the bench scientist".

In this paper we draw on the conceptual insights made by Livingstone in his seminal book on the historical geography of science (Livingstone 2003). Here Livingstone explores how science is made in spaces like the museum, the botanical garden and the coffeehouse, demonstrating how place conditions scientific practices and shapes local identities. He explores the instability of scientific meaning through case studies of how science is received in different places, for example, through a focus on the interactions between science and religion which shaped the production and reception of Darwinian ideas in different places in the nineteenth century (Livingstone 2002, 2003, 2005a, 2005b). Our paper illustrates how Livingstone's geographical perspective illuminates the making of embryonic stem cell science in the twenty-first century.

To do this we utilize Livingstone's (2003, pp. 6–7) argument that: "Both materially and metaphorically the spatial matters a great deal . . . Spaces both enable and constrain discourse [and practice]". Livingstone's geographical analysis of science entails three main elements. First, an exploration of the spatial transformations wrought by science. Second, an investigation of the spatial production, circulation and consumption of science, and third, an examination of the diverse disciplinary spaces of science. Taken together, Livingstone argues that this tripartite geographical approach enables the researcher to "begin to map the major features of modern scientific culture" (Livingstone 2002, p. 32). The three empirical sections of our paper illustrate these three geographical themes and approaches. As far as we are aware, we are the first researchers to employ Livingstone's ideas to examine contemporary laboratory science.

One frequent criticism of Livingstone's research is that it concentrates solely on historical studies of science (Powell 2007). Although this is true, it is also a little unfair as Livingstone has called for more ethnographic research on contemporary science, as we saw above. Another criticism of Livingstone's work is that space as metaphor and place as location are sometimes elided as he views these as two aspects of the same "geographies of science" coin. This is an analytical trap we shall try to avoid by employing Livingstone's distinction between "physical material places" and "relational metaphorical spaces" throughout our paper.

We have discussed the prospects and problems of human embryonic stem (hES) cell research and cell transplantation in a series of STS publications on expectations, ethics and translational research (e.g. Wainwright *et al.* 2006a, 2006b, 2006c, 2007; Michael *et al.* 2007; Cribb *et al.* forthcoming; Williams *et al.* forthcoming). In this paper, however, we employ a "geography of science" approach to the diabetes and hES cell field, applying insights from a Livingstonian geography of science approach to explore what we call stem cell landscapes in the making. We begin by illustrating some of the transformations of the places of stem cell science and medicine. We then move on to a discussion of the influence of place on the production and reception of stem cell science. In the final empirical section of the paper we deconstruct the stem cell transplant approach to diabetes to illuminate some of the ways in which "space matters", and in particular we explore different "disciplinary spaces" in the field of stem cells and diabetes.

Note on methods

In this paper we use data from two ESRC-funded projects on the prospects and problems of translational research in the field of stem cell research. We draw on over 60 in-depth interviews with scientists and clinicians in some of the leading labs and clinics in the UK and the USA, exploring their views on the bench–bedside interface in the fields of neuroscience and diabetes. The interviews lasted between one and two hours, took place within the experts' offices, and with permission were taped and transcribed. Open-ended questions and an informal interview schedule were used, in order to encourage scientists and clinicians to speak in their own words about their experiences. Transcripts were analyzed by content for emergent themes, which were then coded (Strauss 1987). The research team discussed the data and analysis which enabled different perspectives to be incorporated, and added to the richness of the analysis.

Diabetes and stem cells

Diabetes mellitus has been identified as "one of the main threats to human health in the 21st century" (Zimmet *et al.* 2001, p. 782). For example, in the UK 2% of the population have diabetes and treatment accounts for 10% of the total healthcare budget. The current interest in hES cells as a source of insulin-producing β-cells is driven by recent research on curing Type 1 diabetes through human islet cell transplantation (Shapiro *et al.* 2000). The rate-limiting step for this new therapy is the supply of donors, with estimates of around 1000 heart-beating brain-dead donors in the UK per year, against an estimated 100,000 people with Type 1 diabetes. Stem cells are therefore important because they offer, at least in principle, an almost unlimited supply of insulin-producing cells for transplantation (Bonner-Weir and Weir 2005). In the next two sections of the paper we explore some of the ways in which this argument has recently changed the landscape of diabetes and stem cell research.

Transformations of the spaces of stem cell science and medicine

According to Livingstone, the central question of the geography of science is: "Can the location of scientific activity . . . affect the content of science?" (Livingstone 2003, p. 1). As human embryonic stem cells are perhaps the most ethically contested area in contemporary global biomedicine therefore they are an excellent example of the ways in which science reflects geography. For instance, levels of global embryonic stem cell regulation vary from the restrictive (e.g. Italy with an outright ban on hES cell research), through the flexible (e.g. the USA with a very small number of "Presidential hES cell lines" available for government funded research, but almost all research on hES cells allowed if it is privately funded), to the permissive (e.g. the UK – see Salter 2007). Consequently, whether or not hES cell research can be practiced by scientists varies from one country to the next and these variations in what science is allowed to do inevitably shape what Livingstone (2003, p. 23) terms the "sites of speech and locations of locution" of science.

Moreover, Livingstone (2003, p. 124) also contends that scientific knowledge and practice is not only shaped by place, but that they are also important in shaping regional identities. For instance, the San Diego mesa in Southern California was an area of scrubland in the early 1950s but 50 years later this is one of the world's leading centers of bioscience research with the Salk, Scripps and Burnham Institutes recently joining with the University of California San Diego (UCSD) to form the Southern California Institute of Regenerative Medicine. This collaboration is one example of the way in which the California Institute of Regenerative Medicine (CIRM) is changing the global stem cell landscape (Dalton 2005). CIRM's mission is primarily to fund hES cell research in California and their $3 billion of funding over the next decade is a direct response to what is, essentially, President George Bush's ban on Federal funding hES cell research in the USA (Mooney 2005). In other words, place matters and there is, in effect, a global and regional map of where one can practice as a hESC scientist.

Such topographies of hES cell research also reveal "spaces of conversation and spaces of silence" (Livingstone 2002, p. 32). We develop this theme later in the paper in a discussion of the reception and changing expectations of a seminal paper on "Stem cells and diabetes" (Lumelsky *et al.* 2001). However, first we focus on how the geographical transformation of stem cell science was wrought by the seminal Lumelsky *et al.* (2001) paper in *Science*. The transformational potential of Lumelsky *et al.* (2001) was reflected in an accompanying *Science* Editorial which spoke of "alchemy, medical miracles, and hot commodities". We have explored the subsequent unraveling of such claims elsewhere (Michael *et al.* 2007). Here we highlight how this paper, from a NIH lab in the USA, also transformed EU research on stem cells and diabetes through the award of a €2 million grant (2002–2006) designed to enable six labs in Europe to replicate and extend "the Lumelsky protocol".

In fact, these labs have not been able to deliver their "deliverables", namely "the successful construction of bioartificial pancreatic islet micro-organs from glucose-responsive insulin-secreting cells of stem cell origin" (EU Framework V, Cell Factory 2002; see Michael *et al.* 2007). However, our point here is that this EU project used mouse embryonic stem cells, *not* hES cells, because the Italian government would not allow the EU to fund a project on hES cells. This use of mouse rather than human ES cells demonstrates the geographical dictum that: "*Who* we are and, and *where* we are, are interconnected in important ways" (Livingstone 2002, p. 45, emphasis added). In other words, place shapes scientific identity and practice. In the next section we turn to the production and reception of stem cell science.

Geographies of the production and reception of stem cell science

In this section we outline some of the ways in which geographies of scientific consumption also transform the spaces of science, and in so doing, we illustrate Livingstone's claims (2002, p. 30) that "textual meaning is not stable. Texts are read differently in different places." Such a transformation in reading the claims of the Lumelsky protocol is captured in this forthright view:

> You look at that [Lumelsky] paper and see that it is fundamentally flawed from the very beginning. How it passed through the review process I have no idea. When I first read that paper, I thought it was nonsense . . . There was hardly any insulin message, the PDX1 message, which is critical for pancreas and beta cell formation, went *down* when she differentiated her cells . . . It's had an enormous impact, a disastrous impact on the field. I think it's set the field back for a couple of years. And in this country people were reproducing and extending this result . . . It was a terrible thing that happened, as they [scientists] dropped everything and tried to repeat and extend their protocol. And the protocol was wrong. So it's a disaster. How many hundreds of scientific man-years and millions of dollars were devoted to that protocol in one way or another? It's just a complete waste. (Scientist 28, USA)

The Lumelsky paper thus exemplifies a key characteristic of science as outlined by Livingstone who states that:

> Findings [are] always open to suspicion and negotiation . . . Scientific knowing is an inescapably social phenomenon involving judgments about the integrity of people and their practices. Geography makes the scientific enterprise an inescapably moral undertaking. (Livingstone 2003, p. 165)

Here, morals are taken as a means of judging the scientific merits of a paper or line of work as "good or bad science". More broadly, is the testimony of a science paper credible and trustworthy?

Another defining attribute of science is "publish or perish", which means contemporary stem cell science is driven by the need to be productive. We have argued that this requirement for scientific productivity has led to three increasingly radical shifts in the diabetes and hES cell field (Wainwright *et al.* 2007), namely, the cell culture approach (Lumelsky *et al.* 2001); the genetic engineering of stem cells with insulin promotor genes (in 2003); and the use of oncogenes in functioning beta cells (in 2005). Changing the conditions of cell culture as an approach to differentiating ES cells into insulin-producing cells was somewhat disparaged by one of our interviewees:

> A lot of people have used the "shake and bake" approach. They're now learning how to do the molecular approach. (Scientist 25, USA)

This quote is another example of the way in which science can be seen as, in Livingstone's phrase, a "moral undertaking", as molecular biology is implicitly seen as rigorous science while "shake and bake" is seen as less rigorous science.

The second approach entails inserting "insulin promotor genes" (such as PDX1) into hES cells in an attempt to drive these stem cells down a pathway that will produce insulin. This approach was seen as unproblematic by many of the US scientists we interviewed:

> Putting in the PDX genes, that sort of manipulation is fine (Scientist 25, USA)

However, in our UK scientists' views this was seen as problematic because such genetic manipulation to produce functioning cells (for cell transplants to cure diabetes) is colored by the public furore that accompanied the genetic engineering of food crops:

> My guess would be in a society that won't even eat genetically engineered tomatoes that they are not going to let us put genetically engineered cells back into [diabetic] patients we can keep alive perfectly normally. (Scientist 3, UK)

However, so far both the cell culture and insulin promoter gene approaches have failed to make beta cells that produce insulin. The third, and yet more radical approach, essentially starts with insulin-producing beta cells and then genetically engineers these cells with cancer genes to expand the number of functioning cells:

> There's a really interesting article in this month's *Nature Biotechnology* [Narushimi *et al.* 2005], where they used a genetic approach. It was really impressive. What they did was beta cell immortalization. So they took human beta cells, and they put in an oncogene as an immortalization gene. The cells ... were remarkably like *real* beta cells, and they cured diabetes in mice. *I think it's the best anybody has done so far.* Now how reproducible it is? That's a test of time. And then *what is it going to take for someone to have the courage to transplant that into a person?* (Scientist 25, USA, emphasis added)

Here the drive to make cells that produce insulin has persuaded Japanese scientists to develop what many scientists would view as cutting edge but risky science. However, our UK and US scientists questioned whether such a radical move would actually lead to the development of a usable source of transplantable cells. For example, genetically engineering cells with insulin promoter genes is generally viewed as acceptable in the USA, but is resisted in the UK, while using oncogenes is seen as acceptable in Japan but as problematic in both the USA and the UK. In summary, we argue that these shifts in biomedical "spaces of manipulation" (Livingstone 2002, p. 17) from cell culture (in 2001), to insulin genes (in 2003), to oncogenes (in 2005) represent different accounts of what counts as both good science and what are seen as the most promising ways of moving from the lab to the clinic. These temporal shifts are, however, also *spatial* shifts and they can be seen as contemporary examples of what Livingstone (2003, p. 15) describes as "spaces of acceptance and spaces of resistance", as cell biologists in different geographical places argue for or against particular experimental approaches.

In the final section we begin to deconstruct the notion that cell transplantation is the best approach to diabetes. We shift our geographical gaze from the *places* of making beta cells from ES cells to the different disciplinary perspectives afforded by the *spaces* of islet transplantation, immunology, commerce and the hES cell field.

Disciplinary spaces: deconstructing the stem cell transplant approach to diabetes

The "stem cells for cells transplants to cure diabetes" argument is made up of four elements: (1) Islet cell transplants *can* cure Type 1 diabetes; (2) We *need* a new supply of cells (because of the huge shortfall of cadaver cells); (3) ES cells are the *best* source of this unlimited supply cells; Therefore, (4) "ES cell therapy" *will* cure Type 1 diabetes through cell transplantation. In what follows we explore the third element of Livingstone's geographical analysis of science, through what Livingstone refers to as:

> Geographies of the moral economies of epistemic justification . . . Where scientific claims are produced, where scientific knowledge is encountered, and where scientific lives are lived, matter. (Livingstone 2002, pp. 96 and 40).

Hence both physical places and *disciplinary spaces* influence how papers, research programs and the arguments that underlie them are all crucial factors in the social shaping of contested fields within the biomedical sciences. We draw on our interviews with scientists and clinicians to undermine the four steps of the "ES cell therapy will cure diabetes argument" that has been the foundation of the scientific research programs we have discussed so far. Some of our quotations come from clinicians outside the "making beta cells from ES cells" field of science (for example, from the field of clinical islet transplantation), while other quotes represent scientists who are engaged in turning ES cells into beta cells but here they take on, for example, the mantle of immunology to reflect on how their work is seen by scientists from different disciplinary spaces.

First, we examine whether the claim that the prospective future growth of islet transplantation is dependent on an almost unlimited supply of stem cells is reflected by leading clinicians working in the field of islet cell transplantation:

> You live around your diabetes. And to say to someone like that, "We could get rid of this for you", they would sell their souls for that . . . In my opinion it's [cures via cell transplants] a long way away . . . People are saying the islets don't last. And that's the big issue . . . My own feeling is that islet transplantation is a far from efficient procedure, which I would only use *in extremis*. If you use it *in extremis*, the benefits are enormous. (Physician 31, UK)

> We've assessed about 250 patients and we've only got a handful who we felt would be good candidates for this [islet transplantation]. (Surgeon 46, UK)

So it seems that islet transplantation is likely to remain a minor treatment even for those patients with uncontrolled diabetes, as many of these patients' lives can be improved through better medical management. Moreover, longer term follow-up of islet cell transplant patients has shown that the transplanted cells die after some 6–12 months (Gaglia *et al.* 2005).

A second, and in many ways more fundamental threat to the claim that cell transplants are needed to cure diabetes comes from immunology. Type 1 diabetes is an autoimmune disease and therefore transplanting islet cells to produce a cure can be seen as illogical, as the transplanted cells will, in time, be destroyed by the autoimmune disease process that caused a patient's Type 1 diabetes. From this immunological perspective, therefore, no cells are needed for transplants – what is needed instead is a vaccine to cure diabetes:

> As a matter of fact, I think we are going to prevent diabetes before we can cure it by cell replacement . . . My interest in immunology was zero. But that was a big mistake, and we are learning now that if you don't take care of both equally, you will never get cell replacement to work . . . My whole life's work has been on islet cell biology . . . but I believe Type 1 diabetes will be prevented through the identification of the antigen, and through vaccines. (Scientist 49, USA)

A third threat to the growth of cell transplant approaches comes from the extremely powerful diabetes industry. Cell transplants are seen as a threat to the future profitability of the many large biomedical manufactures and pharmaceutical companies who dominate current treatments for diabetes:

> If you can turn cells into beta cells and transplant them there are an awful lot of drug companies that would be very unhappy with that. (Scientist 3, UK)

Moreover, the development of sophisticated (and expensive) insulin pump regimes – which continually sense the patient's blood sugar level and then titrate the delivery of the appropriate amount of insulin – is a major rival to the development of islet cell transplants:

> There's a race . . . you either believe that . . . what the diabetic needs is islet cells. Or you say, "Well we're great scientists, what we could do is put together the perfect glucose sensor linked to the perfect insulin delivery device, and away you go." Now they are getting closer to that . . . Britain and the NHS are too cheap for the companies who make them to be that interested. They'd rather sell these things to the Americans, where the rich can buy them and the poor can go hang. (Physician 31, UK)

Such differences illustrate Livingstone's (2003, p. 11). point that: "As ideas circulate, they undergo translation and transformation because people encounter representations differently in different circumstances". The three examples we have examined so far – from the perspectives of islet transplant practitioners, diabetes immunologists and the diabetes industry – are all examples of the competing sub-fields of translational research on diabetes. Our fourth example, however, shifts the focus to how scientists elsewhere in the field of hES cell research view the whole enterprise of stem cells as the engine for the growth of cell transplantation to cure chronic disease. Within the hES cell field another way for stem cell translation to occur is by using hES cells to study the genetics of a "disease in a dish" in order to develop pharmacological therapies (Scott 2006). The argument for a paradigm shift to "disease in a dish" approaches was eloquently made by a US stem cell scientist:

> Stem cells have been, over the last two or three years, sold to the public as a potential therapeutic application for transplantation . . . And I think some opportunists have jumped into this field, done some rat studies with human ES cells and some changes occurred. I think

> that people were shaken and some scientists started backing off and saying it's all hype, there are no real cures in this domain . . . For the last year or so I've been talking about how you can study "diseases in a dish" through cell culture. This is a revolution in human biology. This is a paradigm shift . . . This is going to happen. It's too clear. It's too right. (Scientist 29, USA)

In summary, we have examined four ways in which the field of "ES cell therapy as a cure for diabetes" may be undermined by exploring different disciplinary spaces. In other words, we have illustrated how the research program to use hES cells to make insulin-producing cells for cell transplants is consumed depends on the disciplinary spaces of geographies of reading science (Livingstone 2005a). Drawing on Livingstone (2003, p. 4) we argue that, "as it moves it is modified, as it travels it transforms". Perhaps the stem cells for cell transplants field represents an imagined future geography of a new world of regenerative medicine?

Discussion and conclusion

We have explored the social, medical and scientific aspects of new medical technologies as an exemplar of the making and the shaping of contested biomedical landscapes. We argue that Livingstone's geography of science provides a novel and productive approach for understanding "the evolution of a revolution" in the emerging field of regenerative medicine. We agree with Livingstone's (2002, p. 85) claim that "ethnographic investigations of the cultures of contemporary [scientific] geographical practices, seems a very fertile and exciting line of research". Hence we hope our text is testimony that spaces of speech and places of performance are vital features of analysis of the field of stem cell science.

Geographical metaphors play an important role in both science and science studies (Glasner 1998). Scientists cross boundaries, map new areas, chart topographies, trace contours, transform landscapes, and experience highs and lows as they explore new paths that lead from the panoramic vistas of mountain peaks to the swampy lowlands of the valleys below. The scientific field itself is a terrain of tectonic shifts, backwaters and hotspots. A vivid example of the way in which scientists think of the uneven distribution of such scientific capital was given to us by an eminent scientist (a Fellow of the Royal Society) who asserted that in UK science: "there's London, Oxford and Cambridge – and then it's a desert until you reach Edinburgh!" (Scientist 13, UK). Clearly, geography matters! However, as we saw in our introduction, STS research (and we include all of our own publications on new medical technologies here) privileges the social over the spatial. Just how much geographical perspectives matter in understanding the world is articulated by David Livingstone:

> Cultivating a geography of science will disclose how scientific knowledge bears the imprint of its location . . . Attending to the spaces of science is of no small significance in coming to grips with the character of scientific endeavour and therefore of the modern world itself. (Livingstone 2003, pp. 13 and 186)

What we have attempted to do, therefore, is to use three specific insights from Livingstone's historical writings on nineteenth-century science to construct a geographical analysis of our ethnographic data on twenty-first-century stem cell science.

In this paper we have investigated the interrelated fields of diabetes and hES cell research and we have explored some of the ways in which time, and particularly place, color the interactions between expectations, science and society. Science entails a struggle for disciplinary futures. We have shown how, in areas of "science-in-the-making" such as stem cell research, what counts as credible, trustworthy and productive science is highly contested in what Livingstone (2002, p. 96) calls "moral economies of epistemic justification". We realize that the phrase "moral economies" has a variety of meanings (for instance Daston 1995, Salter 2007). However, we employ the phrase to capture how science entails an economy of production, circulation and consumption of scientific goods, and that scientists engaged in this economy inevitably have to make "moral decisions" about what constitutes credible, trustworthy and justifiable research or, to put it more boldly, good and bad science.

Moreover, we trust that we have both evoked the revolutionary claims that are being made by stem cell scientists, and also reflected some of the "anti-hype" that many scientists and clinicians also express about the field of stem cell research. Our case studies and examples illustrate the geographical thesis that: "Where scientific texts are read has an important bearing on how they are read" (Livingstone 2005a, p. 391). The where here refers, of course, to both "physical material places" and "relational metaphorical spaces". We hope our paper illustrates some of the ways in which geographies of science provide a fruitful approach to understanding the making of the contested landscapes of biomedicine.

Acknowledgements

This paper is based on two research projects: ESRC Stem Cell Initiative RES-340-25-0003 and RES-350-27-0001. We would like to thank both the Editors of this special issue and our three anonymous referees for their valuable help in developing this paper.

References

Barnes, T.J., 2001. In the beginning was economic geography: a science studies approach to disciplinary history. *Progress in Human Geography*, 27, 69–95.

Bonner-Weir, S. and Weir, G.C., 2005. New sources of pancreatic beta-cells. *Nature Biotechnology*, 23, 857–886.

Cribb, A., *et al.*, forthcoming. Towards the applied: the construction of ethical positions in stem cell translational research. *Medicine, Health Care & Philosophy.*

Dalton, R., 2005. California prepares to roll out stem-cell funding. *Nature*, 7060, 800–801.

Daston, L., 1995. The moral economy of science. *Osiris*, 10, 2–24.

Gaglia, J.L., Shapiro, A.M.J., and Weir, G.C., 2005. Islet transplantation: progress and challenge. *Archives of Medical Research*, 36, 273–280.

Gieryn, T.F., 1999. *Cultural boundaries of science: credibility on the line.* Chicago University Press.
Glasner, P., 1998. Patrolling the boundaries or tracing the contours: cartographic metaphors and the human genome project. *In*: P. Wheale, R. von Schomberg, R., and P. Glasner, eds. *The social management of genetic engineering*. Aldershot: Ashgate, 29–44.
Henke, C.R. and Gieryn, T.F., 2008. Sites of scientific practice: the enduring imporatance of place. *In*: E.J. Hackett, *et al.*, eds. *The handbook of science and technology studies*. 3rd ed. Cambridge: MIT Press, 353–376.
Johnstone, R.J. and Sideway, J.D., 2004. *Geography and geographers: Anglo-American human geography since 1945*. 6th ed. London: Hodder Arnold.
Latour, B., 1988. *The pasteurization of France.* Cambridge, MA: Harvard University Press.
Latour, B. and Woolgar, S., 1986. *Laboratory life: the construction of scientific facts*. 2nd ed. Princeton University Press.
Livingstone, D.N., 2002. Science, space and hermeneutics. The Hettner Lectures 2001. Heidelberg: University of Heidelberg.
Livingstone, D.N., 2003. *Putting science in its place: geographies of scientific knowledge*. Chicago University Press.
Livingstone, D.N., 2005a. Science, text and space: thoughts on the geography of reading. *Transactions of the Institute of British Geographers*, 30 (4), 391–401.
Livingstone, D.N., 2005b. Text, talk and testimony: geographical reflections on scientific habits. *British Journal for the History of Science*, 38, 93–100.
Lumelsky, N., *et al.*, 2001. Differentiation of embryonic stem cells to insulin-secreting structures similar to pancreatic islets. *Science*, 292, 1389–1394.
Michael, M., *et al.*, 2007. From core set to assemblage: on the dynamics of exclusion and inclusion in the failure to derive beta cells from embryonic stem cells. *Science Studies*, 20 (1), 5–25.
Mooney, C., 2005. *The Republican war on science.* New York: Basic Books.
Narushima, M., *et al.*, 2005. A human β-cell line for transplantation therapy to control type 1 diabetes. *Nature Biotechnology*, 23, 1274–1282.
Popper, K., 1959. *The logic of scientific discovery.* London: Unwin Hyman.
Powell, R.C., 2007. Geographies of science: histories, localities, practices, futures. *Progress in Human Geography*, 31, 309–329.
Salter, B., 2007. Bioethics and the global moral economy: the cultural politics of human embryonic stem cell science. *Science, Technology & Human Values*, 32 (5), 1–28.
Scott, C.T., 2006. *Stem cell now: from the experiment that shook the world to the new politics of life*. New York: Pi Press.
Shapin, S., 1998. Placing the view from nowhere: historical and sociological problems in the location of science. *Transactions of the Institute of British Geographers*, 23, 1–8.
Shapin, S. and Shaffer, S., 1985. *Leviathan and the air pump: Hobbes, Boyle and the experimental life*. Princeton University Press.
Shapiro, A.M.J., *et al.*, 2000. Islet transplantation in seven patients with Type 1 diabetes mellitus using a glucocorticoid-free immunosuppressive regime. *New England Journal of Medicine*, 343, 230–238.
Strauss, A.L., 1987. *Qualitative analysis for social scientists*. Cambridge University Press.
Wainwright, S.P., *et al.*, 2006a. From bench to bedside? Biomedical scientists' expectations of stem cell science as a future therapy for diabetes. *Social Science & Medicine*, 63, 2052–2064.
Wainwright, S.P., *et al.*, 2006b. Ethical boundary work in the embryonic stem cell laboratory. *Sociology of Health & Illness*, 28, 732–748.
Wainwright, S.P., *et al.*, 2006c. Real science, biological bodies and stem cells: constructing images of beta cells in the biomedical science lab. *Social Theory & Health*, 4, 275–298.

Wainwright, S.P., *et al.*, 2007. Remaking the body? Scientists' genetic discourses and practices as examples of changing expectations on embryonic stem cell therapy for diabetes. *New Genetics and Society*, 26, 251–268.

Williams, C., *et al.*, forthcoming. Human embryos as boundary objects? Some reflections on the biomedical worlds of embryonic stem cells and pre-implantation genetic diagnosis. *New Genetics and Society*.

Zimmet, P., Alberti, K.G.M.M., and Shaw, J., 2001. Global and societal implications of the diabetes epidemic. *Nature*, 414, 782–787.